

Course Code: BCE-O633
Course Name: INTRODUCTION TO DATA SCIENCE AND DESIGN THINKING

MM: 100 Time: 3 Hr. L T P 3 0 0	Sessional:30 ESE:70 Credit :3
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Prerequisites:	Elementary programming knowledge
Objectives:	• An understanding of problems solvable with data science and an ability to attack them from a statistical perspective. • An understanding of when to use supervised and unsupervised statistical learning methods on labeled and unlabeled data-rich problems. • The ability to create data analytical pipelines and applications in Python. • Familiarity with the Python data science ecosystem and the various tools needed to continue developing as a data scientist. • To learn the basics of design thinking and good design concepts • To explore design thinking applications in computer science • To understand design-based issues of product and services • Demonstrate the value of developing a local network and assist students in making lasting connections with the business community. • Students develop a portfolio of work to set them apart in the job market. • Provide an authentic opportunity for students to develop teamwork and leadership skills.

NOTE:	The question paper shall consist of two sections A and B. Section A contains 10 short type questions of 6 marks each and student shall be required to attempt any five questions. Section B contains 8 long type questions of ten marks each and student shall be required to attempt any four questions. Questions shall be uniformly distributed from the entire syllabus.
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Module	Course Content	No. of Hours	POs mapped	PSOs mapped
Module-1	Introduction to Data Science –Evolution of Data Science – Data Science Roles – Stages in a Data Science Project – Applications of Data Science in various fields – Data Security Issues.	09	PO1/ PO2/ PO12	PSO1/ PSO2
Module-2	Data Collection Strategies –Data Pre-Processing Overview – Data Cleaning – Data Integration and Transformation – Data Reduction – Data Discretization.	09	PO2/ PO4/ PO5	PSO1/ PSO2
Module-3	Descriptive Statistics – Mean, Standard Deviation, Skewness and Kurtosis – Box Plots – Pivot Table – Heat Map – Correlation Statistics – ANOVA.	07	PO1/ PO2/ PO3/ PO5	PSO1/ PSO2
Module-4	Introduction to Design Thinking: Definition of design thinking, good design and bad design, importance of design thinking, applications of design thinking. Stages of Design thinking: Empathize, Define, Ideate, Prototype, Test and Implement. The evolution of technology using design thinking, innovative examples of design thinking - Life Saving Dot, Embrace Incubator, Project Bloks, Pillpack, Aarambh Desk.	07	PO1/ PO3/ PO4/ PO5/ PO11	PSO1/ PSO2
Module-5	Case Studies, Design for Specific Culture: Case studies of Zip line, Tesla, AirBNB, The body Shop, Patagonia, Ben & Jerry's, 23 and Me, War child, Warby Parker and Toms Shoes.	08	PO4/ PO5/ PO12	PSO1/ PSO2
Total No. of Hours		40		

Learning Outcomes:	• Demonstrate proficiency with statistical analysis of data. • Demonstrate skill in data management. • Apply data science concepts and methods to solve problems in real-world. • Develop a strong understanding of the design process • Learn how to create physical prototypes / a visual representation of an idea
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Suggested books:

S. No.	Name of Authors /Books /Publisher/Year
1.	Saltz, Jeffrey S., and Jeffrey M. Stanton. <i>An introduction to data science</i> . Sage Publications, 2017.
2.	JojoMoolayil, "Smarter Decisions: The Intersection of IoT and Data Science", PACKT, 2016.
3.	Cathy O'Neil and Rachel Schutt, "Doing Data Science", O'Reilly, 2015.
4.	Luchs, Michael G. "A brief introduction to design thinking." <i>Design thinking: New product development essentials from the PDMA</i> (2015): 1-12.
5.	Suyash Bhardwaj, "10 Amazing Stories of Design Thinking that Shaped the Future: Learning Through Design Thinking", Amazon, 2023. ISBN - 978-93-5906-723-0
6.	David Dietrich, Barry Heller, Beibei Yang, "Data Science and Big data Analytics", EMC 2013
7.	Raj, Pethuru, "Handbook of Research on Cloud Infrastructures for Big Data Analytics", IGI Global.
8.	Cross, Nigel. <i>Design thinking: Understanding how designers think and work</i> . Berg, 2011.
9.	Meinel, Christoph, and Larry Leifer. "Design thinking research." <i>Design thinking research</i> . Springer, Berlin, Heidelberg, 2012. 1-11.

CO-PO/PSO MAPPING														
Course Outcomes (COs)	Program Outcomes (POs)												Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	✓	✓											✓	✓
CO2	✓	✓	✓								✓		✓	✓
CO3	✓	✓	✓	✓								✓	✓	✓
CO4	✓	✓		✓	✓							✓	✓	✓
CO5	✓	✓	✓	✓	✓							✓	✓	✓